



CIS 678 Machine Learning

Special Topics

Course Review Week (Part 2)



Overview Data Code Models Discussion Leaderboard Rules Team Submissions Settings

11

Competition Host
Kamrul Hasan

Prizes & Awards
Kudos [Edit](#)
Does not award Points or Medals

Prizes & Awards
Kudos [Edit](#)
Does not award Points or Medals

Participation
57 Entrants
57 Participants
57 Teams
413 Submissions

Tags

[illegible]



Convolutional NNs

- State of the Art for CV and some other problems
- Filters/Convolutional Kernels

Examples:

- *Alexnet*
- *VGG*
- *ResNet*
- *GoogLeNet*
- ..



2D Convolution

Data matrix (D):

1	0	1	1	0
1	0	0	1	1
0	1	0	0	0
1	0	0	0	1
0	1	0	1	0

Kernel matrix (D):

0	0	0
0	5	0
0	0	0

2D Convolution

Data matrix (D):

1	0	1	1	0
1	0	0	1	1
0	0	0	0	0
1	0	0	0	1
0	1	0	1	0



Output Matrix

0		

2D Convolution

Data matrix (D):

1	0	1	1	0
1	0	0	1	1
0	1	0	0	0
1	0	0	0	1
0	1	0	1	0



Output Matrix

0	0	

2D Convolution

Data matrix (D):

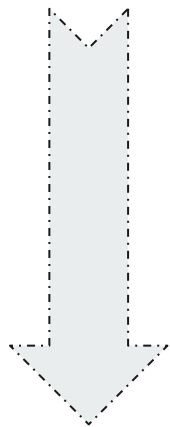
1	0	1	1	0
1	0	0	5	1
0	1	0	0	0
1	0	0	0	1
0	1	0	1	0



Output Matrix

0	0	5

2D Convolution



Data matrix (D):

1	0	1	1	0
1	0	0	1	1
0	5	0	0	0
1	0	0	0	1
0	1	0	1	0



Output Matrix

0	0	5
5		



2D Convolution

Data matrix (D):

1	0	1	1	0
1	0	0	1	1
0	1	0	0	0
1	0	0	0	1
0	1	0	1	0

Kernel matrix (D):

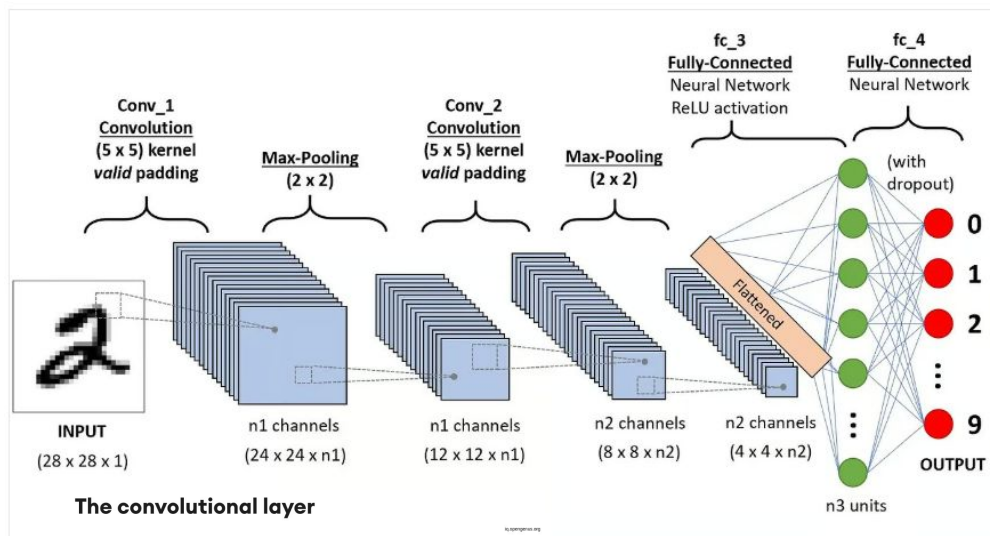
0	0	0
0	5	0
0	0	0

Output Matrix

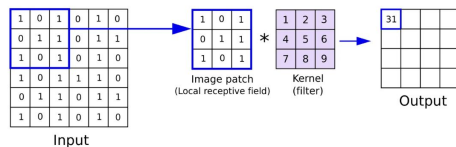
0	0	5
5	0	0
0	0	0



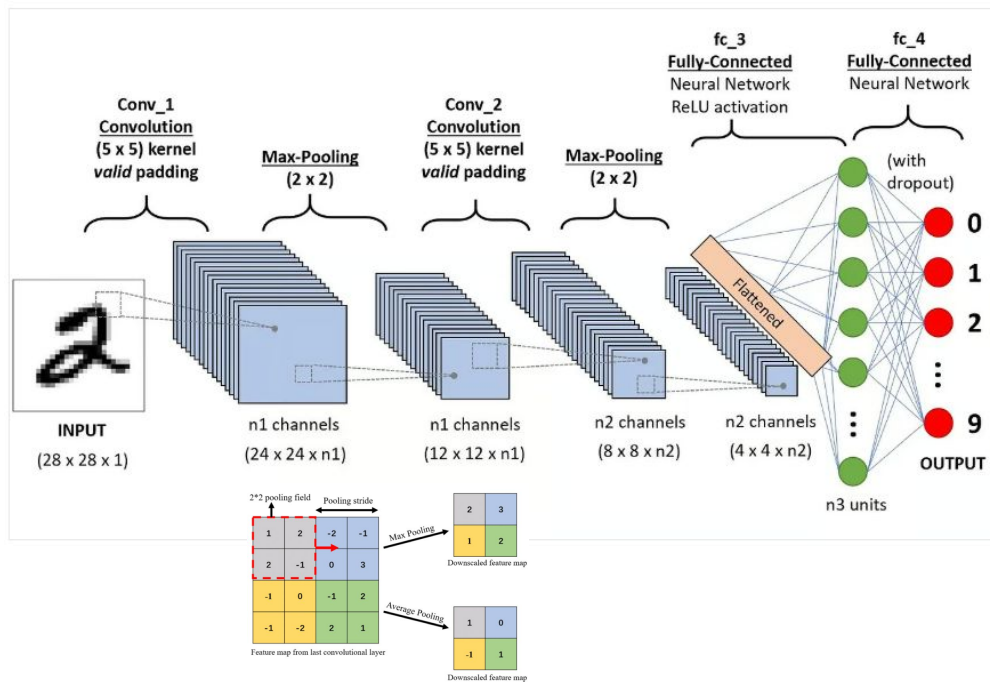
Convolutional NNs



The convolutional layer



Convolutional NNs





Overview Data Code Models Discussion Leaderboard Rules Team Submissions Settings

11

Competition Host
Kamrul Hasan

Prizes & Awards
Kudos [Edit](#)
Does not award Points or Medals

Prizes & Awards
Kudos [Edit](#)
Does not award Points or Medals

Participation
57 Entrants
57 Participants
57 Teams
413 Submissions

Tags

Close
12 days to go





[illegible]



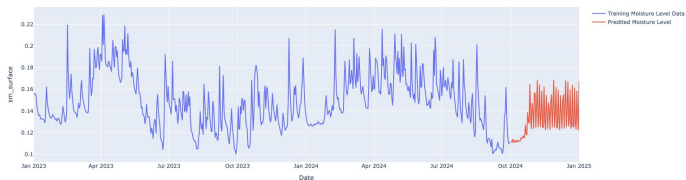
Competition Host
Kamrul Hasan

Prizes & Awards

Does not award Points or Medals

Participation
56 Entrants
53 Participants
53 Teams
572 Submissions

Close

[illegible]

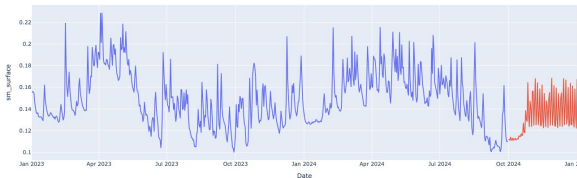


Competition Host
Kamrul Hasan

Prizes & Awards

Does not award Points or Medals

53 Participants
53 Teams
572 Submissions



— Training Moisture Level Data
— Predicted Moisture Level

[illegible]



How are the time-series problems different?

- The models (Regression and Classification), we have learned so far are of the form:

$$f(y|X)$$

- Purely time series models are of form :

$$f(y_t | y_{t-1}, y_{t-2}, \dots, y_0), \text{ where there is no explicit, } X.$$

- For certain data, especially the time series, we can take advantage of the form:

$$f(y_t | X, y_{t-1}, y_{t-2}, \dots, y_0); \text{ essentially, here the input features are } X \text{ plus the lagged instances of the target } y.$$



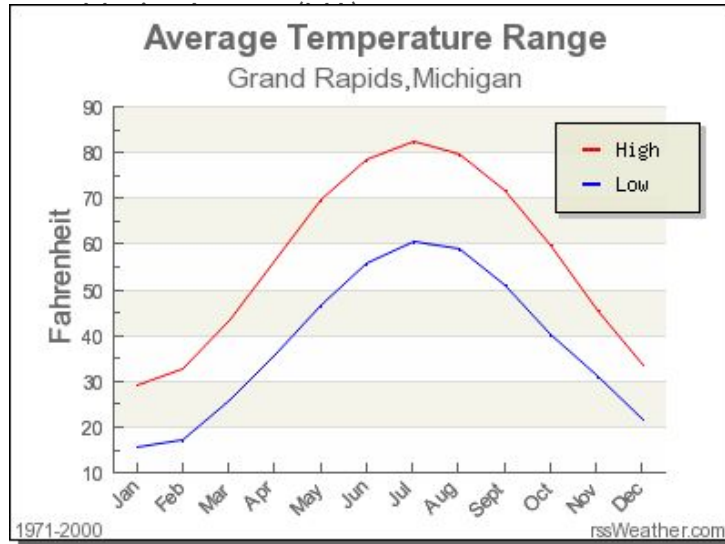
Statistical time series models

- Autoregressive Models (AR)
- Moving Average (MA)
- Autoregressive Moving Average (ARMA)
- Autoregressive Moving Integrated Average (ARIMA)

$$f(y_t | y_{t-1}, y_{t-2}, \dots, y_0)$$

Statistical time series models

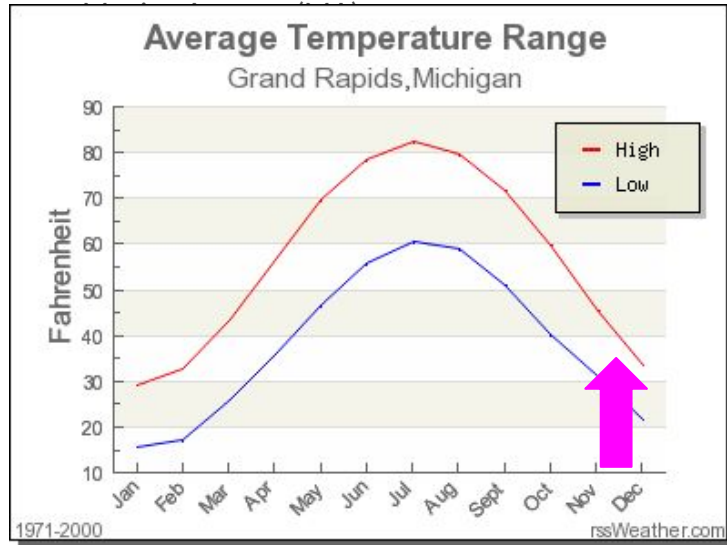
- Autoregressive Models (AR)



$$f(y_t | y_{t-1}, y_{t-2}, \dots, y_0)$$

Statistical time series models

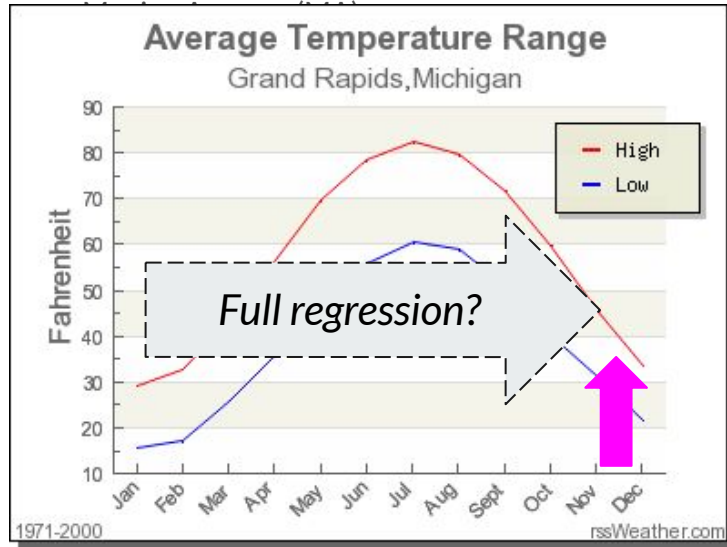
- Autoregressive Models (AR)



$$f(y_t | y_{t-1}, y_{t-2}, \dots, y_0)$$

Statistical time series models

- Autoregressive Models (AR)

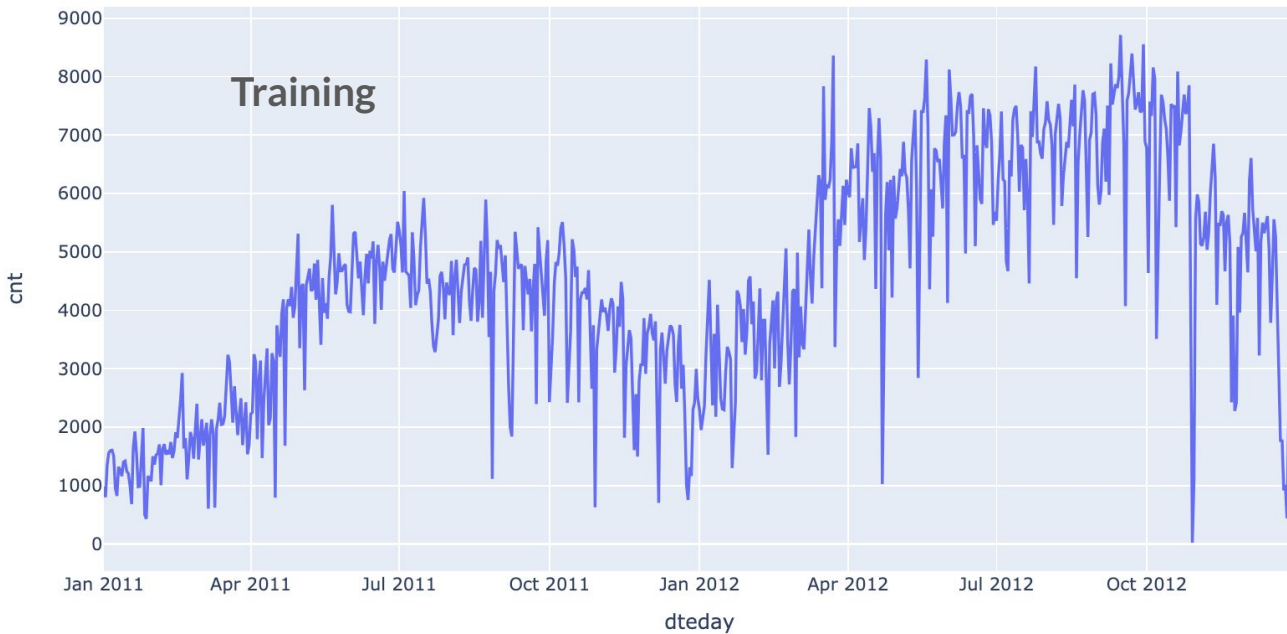


$$f(y_t | y_{t-1}, y_{t-2}, \dots, y_0)$$



How to CV Sequential Data/Models

How to CV Sequential Data/Models



*Always
follow!*

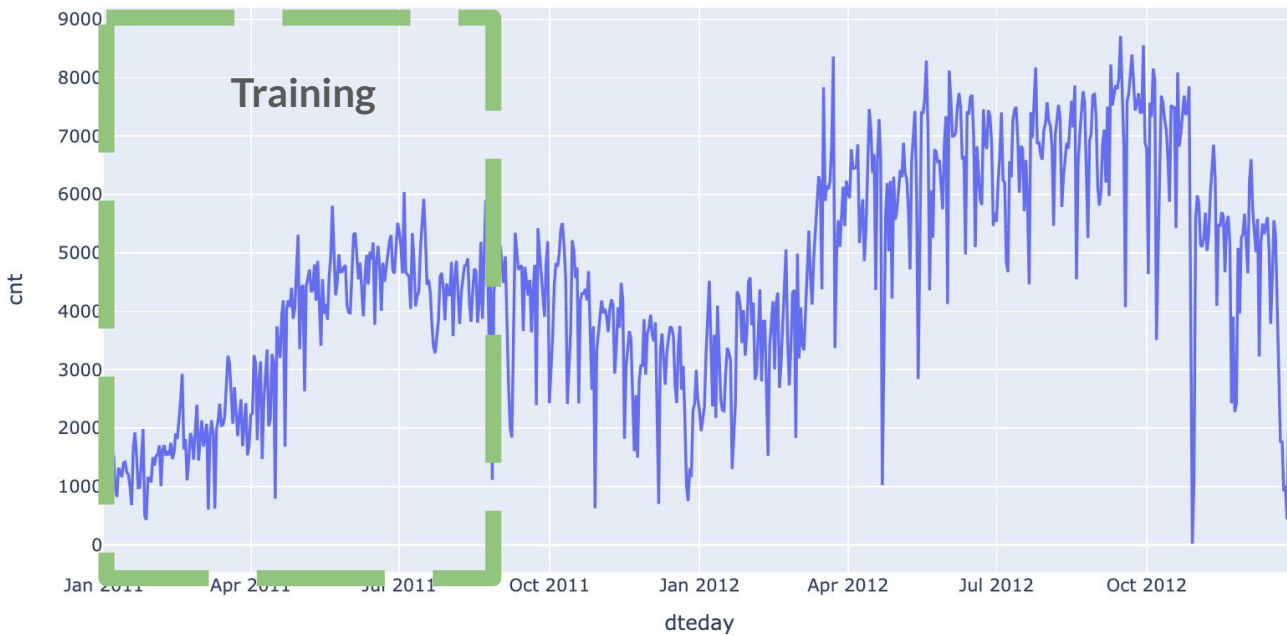
Sequential cross validation



*Invalid
configuration:*

*Training on the last
two and validation
on the **first fold**.*

Sequential cross validation



Always follow!

Sequential cross validation



Always follow!

Sequential cross validation



Always follow!

Sequential cross validation



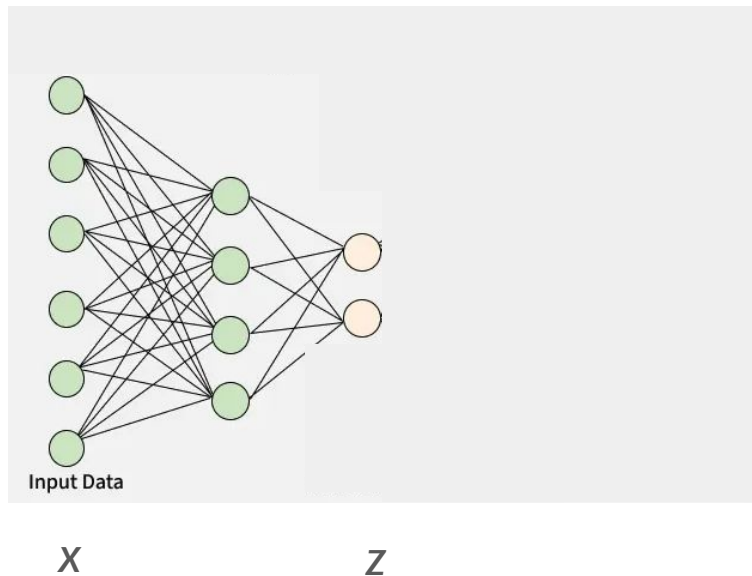
Always follow!



[illegible]

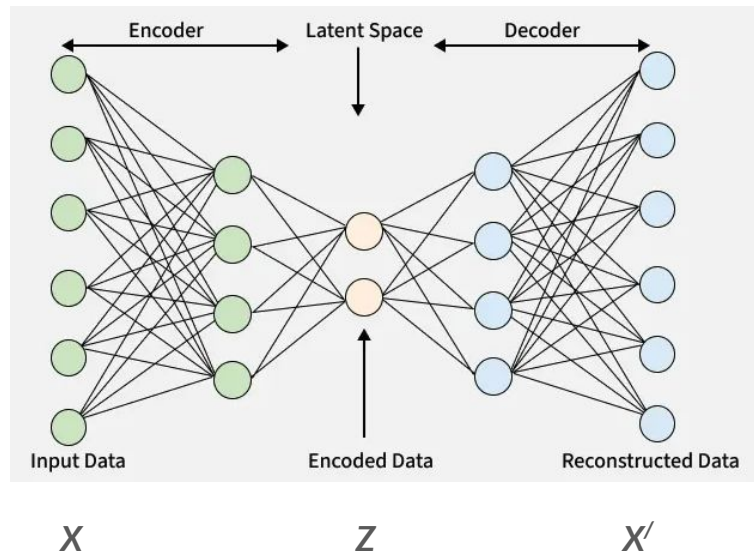
Unsupervised learning (nonlinear)

- Auto Encoders
- Restricted Boltzmann Machines (RBMs)



Unsupervised learning (nonlinear)

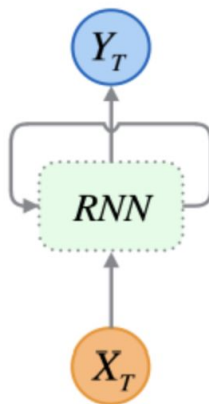
- Auto Encoders
- Restricted Boltzmann Machines (RBMs)



Recurrent Neural Networks

Originally designed to Solve:

- Sequential Problems
 - o *NLP (MT, ...), Speech Recognition,...*
 - o *DNA Sequencing*
- Time-series Problems

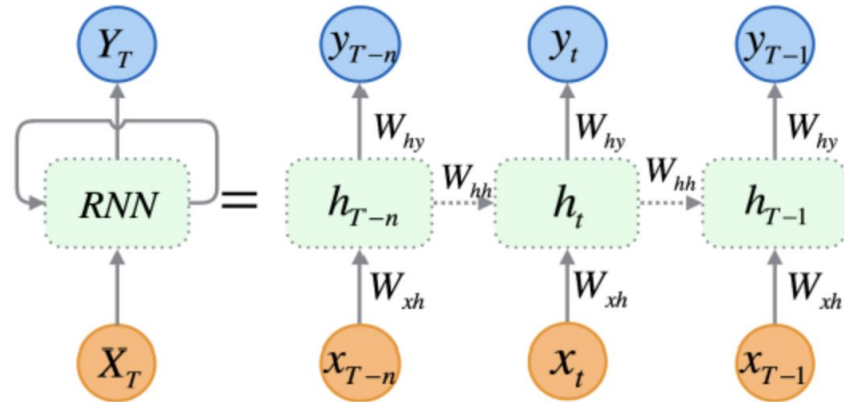


[ref](#)

Recurrent Neural Networks

Originally designed to Solve:

- Sequential Problems
 - o NLP (MT, ...), Speech Recognition,...
 - o DNA Sequencing
- Time-series Problems



Transformers

Examples:

- Encoder decoder pair
- BERT Encoder
- GPT Decoder

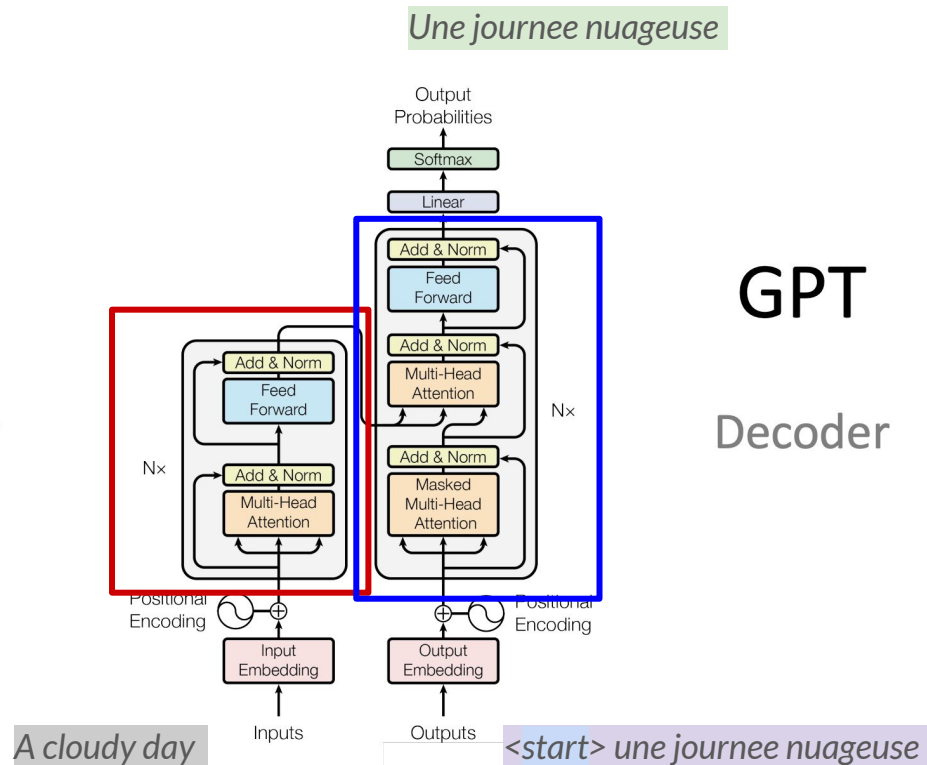
Example Task: **Machine Translation**

BERT

Encoder

GPT

Decoder



Transformers

Examples:

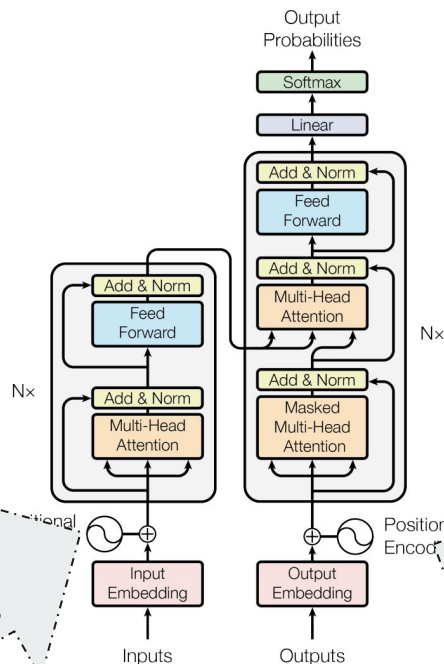
- Encoder decoder pair
- BERT Encoder
- GPT Decoder

A cloudy day

Day by day, the weather got cloudier.

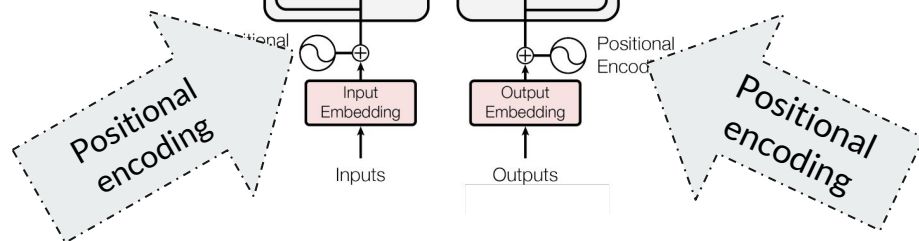
BERT

Encoder



GPT

Decoder



Transformers

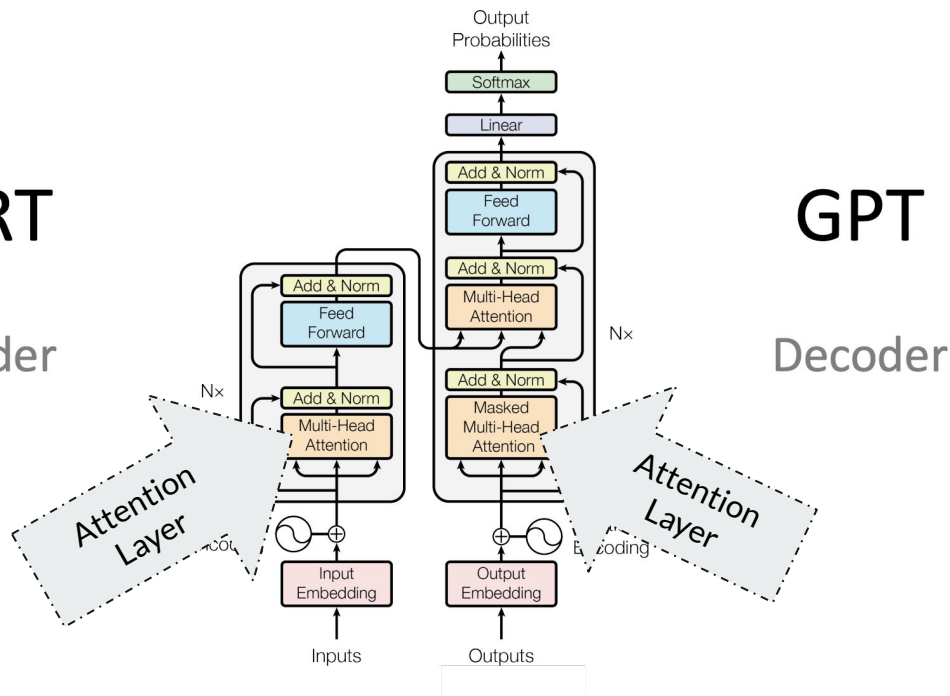
Examples:

- Encoder decoder pair
- BERT Encoder
- GPT Decoder

The animal didn't cross the street because it was too tired

BERT

Encoder



Transformers

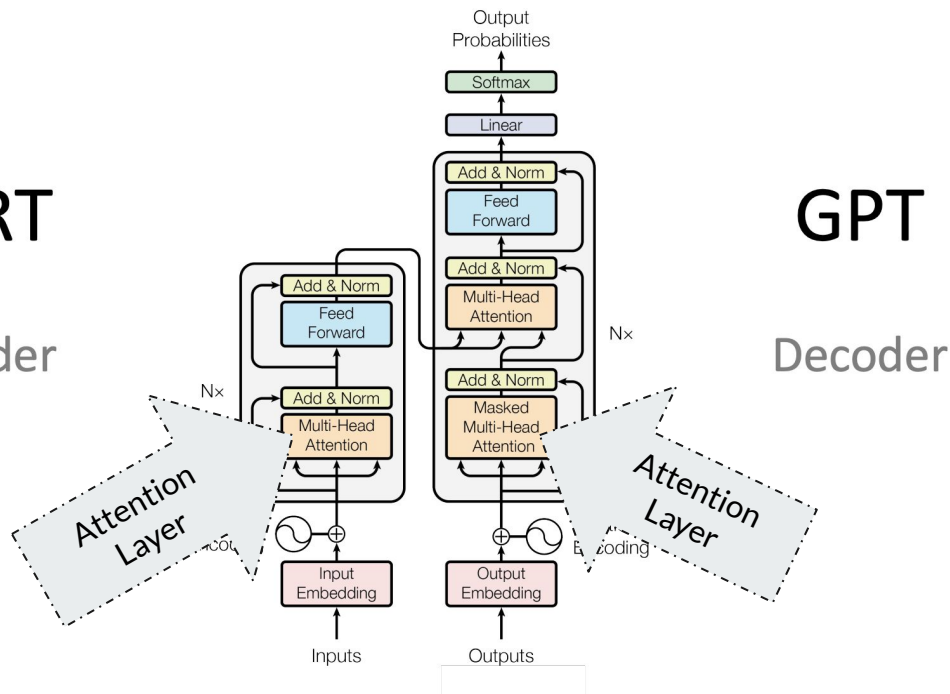
Examples:

- Encoder decoder pair
- BERT Encoder
- GPT Decoder

The animal didn't cross the street because it was too tired

BERT

Encoder



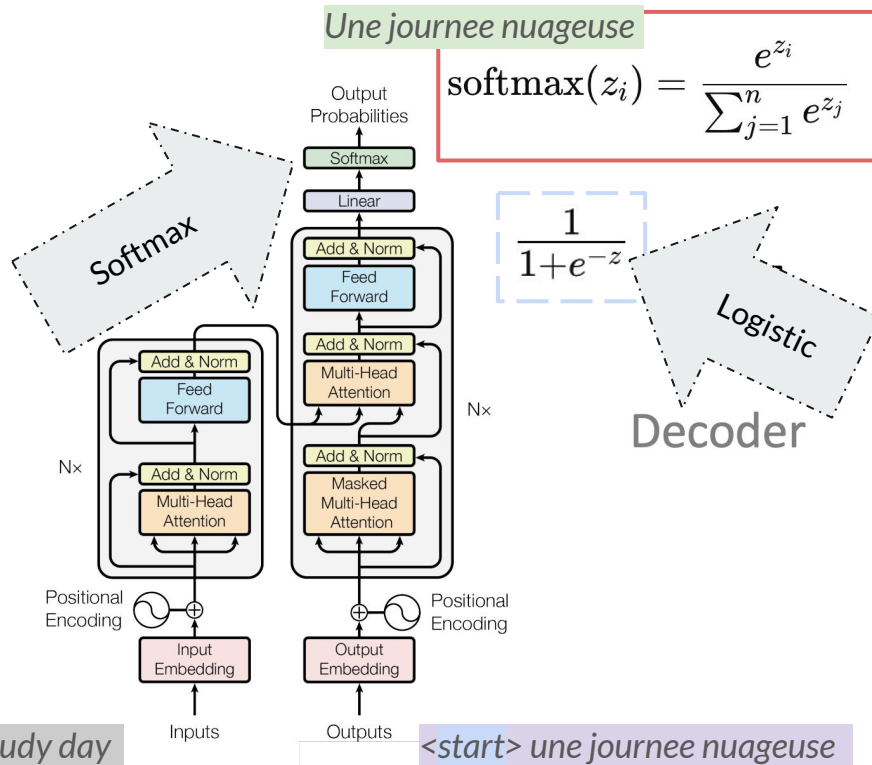
Transformers

Examples:

- Encoder decoder pair
- BERT Encoder
- GPT Decoder

Example Task: Machine Translation

BERT
Encoder



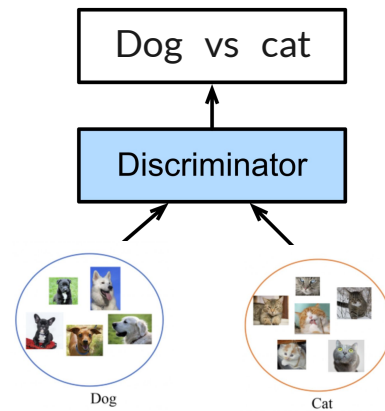


Generative AI

Examples:

- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)

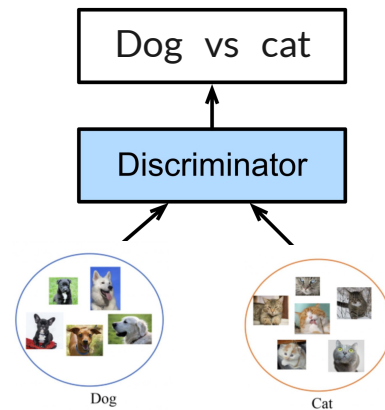
Generative AI



Generative AI

Examples:

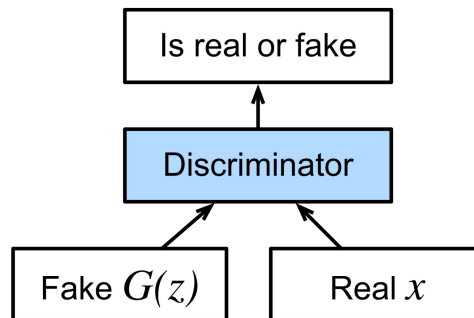
- Generative Adversarial Networks (GANs)



Generative AI

Examples:

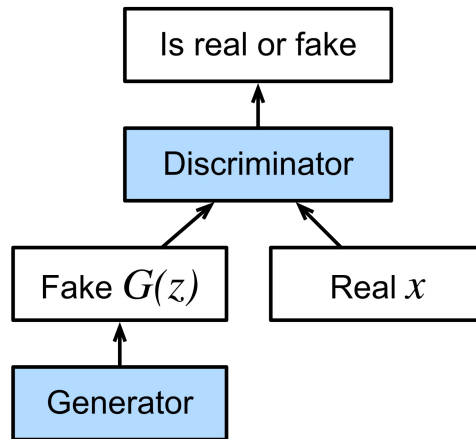
- Generative Adversarial Networks (GANs)



Generative AI

Examples:

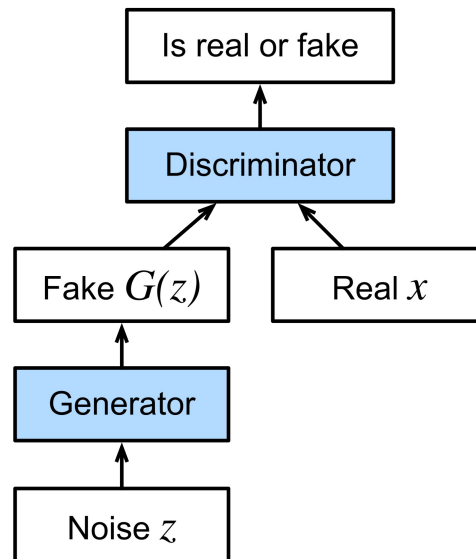
- Generative Adversarial Networks (GANs)



Generative AI

Examples:

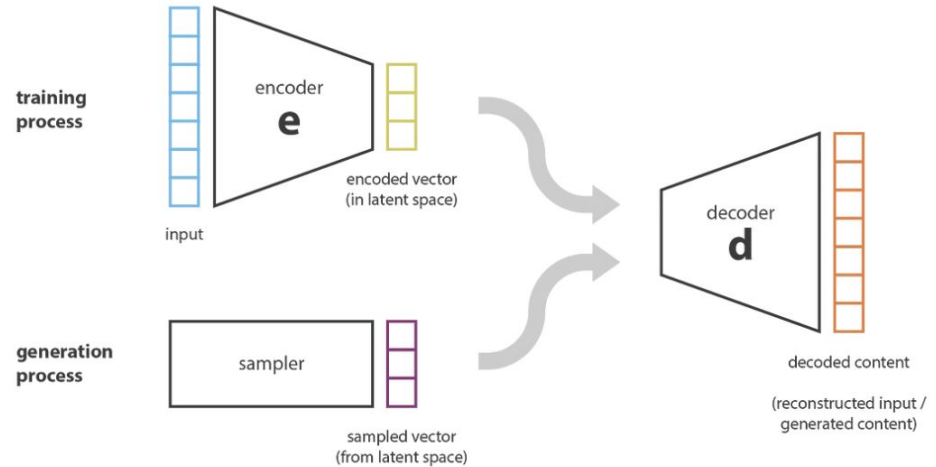
- Generative Adversarial Networks (GANs)



Generative AI

Examples:

- Generative Adversarial Networks (G
- **Variational Autoencoders (VAEs)**

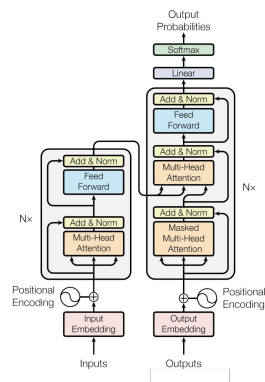


Foundational models

- We don't train independent models explicitly (say machine translation)

BERT

Encoder



GPT

Decoder

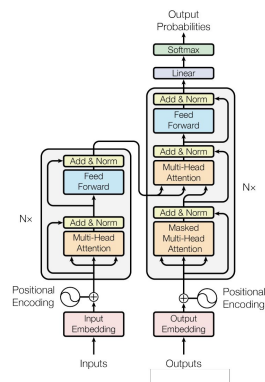
Machine translation

Foundational models

- We don't train independent models explicitly (say machine translation)
- Or document summarization

BERT

Encoder



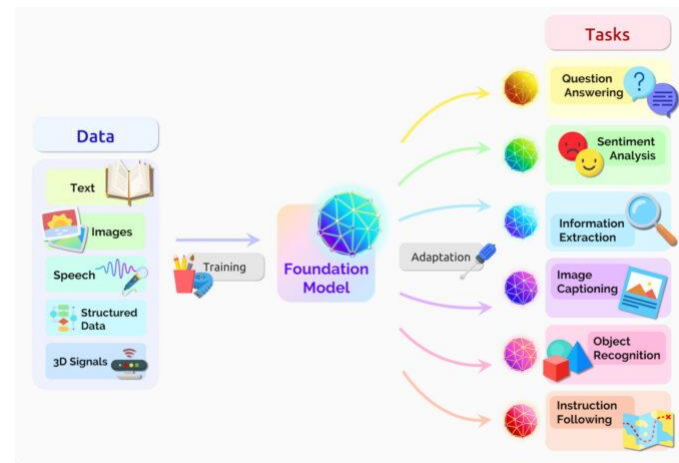
GPT

Decoder

Document summarization

Foundational models

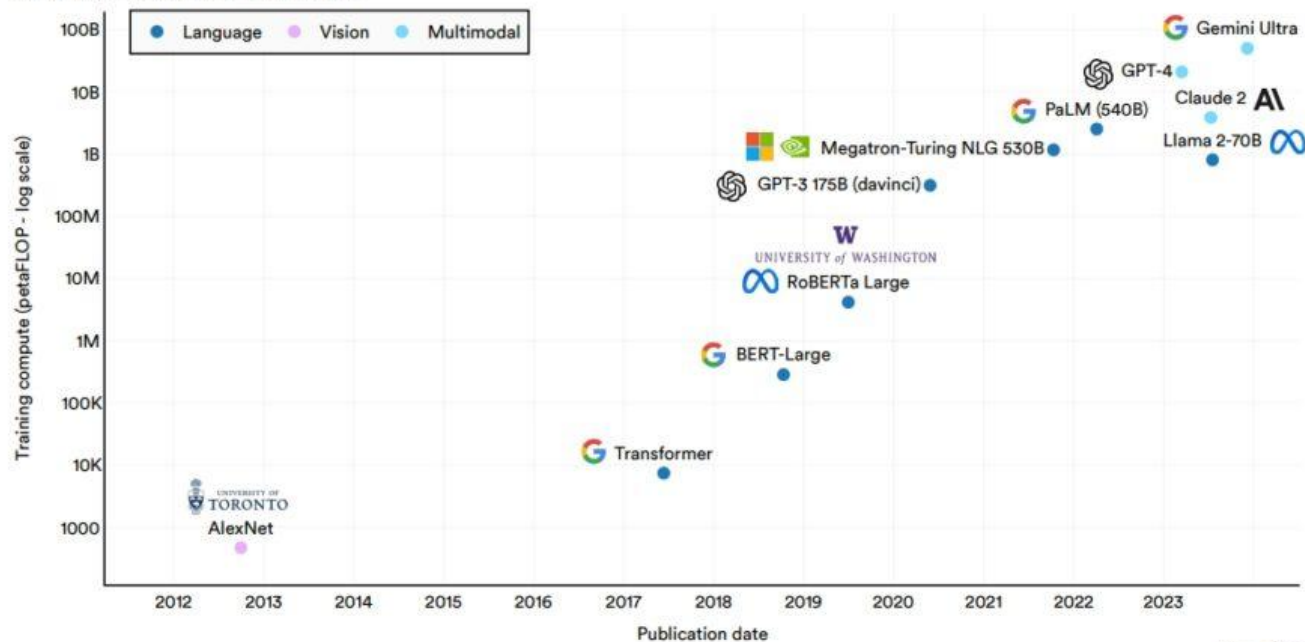
- We don't train independent models explicitly (say machine translation)
- Or document summarization
- **We train a Base model for multiple tasks jointly, and then fine tune for specific tasks**



Foundational models

Training compute of notable machine learning models by domain, 2012–23

Source: Epoch, 2023 | Chart: 2024 AI Index report





Training Neural Network Challenges

- Intractable gradients
 - Vanishing, and
 - Exploding gradients



Training Neural Network Challenges

- Intractable gradients
 - Vanishing, and
 - Exploding gradients
- Various normalizations
 - Input normalization (standard scalar)
 - Batch normalization
 - Layer normalization



Training Neural Network Challenges

- Intractable gradients
 - Vanishing, and
 - Exploding gradients
- Various normalizations
 - Input normalization (standard scalar)
 - Batch normalization
 - Layer normalization
- Controlling overfitting
 - Regularization
 - Early stopping
 - Drop out



QA